Arabic Sentiment Stability Under Paraphrasing: A Comparative Evaluation of Gemini, GPT-OSS, and AraBERT

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***Abstract*—Arabic sentiment analysis models are often evaluated using accuracy-based metrics, but less attention is given to whether their predictions remain stable when the input text is paraphrased. This project investigates sentiment stability under paraphrasing using a combined Arabic Twitter dataset built from ASTD and ArSarcasm. A balanced sample of 150 tweets was paraphrased using Gemini 2.5 Flash, and three models were evaluated on both the original and paraphrased texts: Gemini 2.5 Flash, GPT-OSS-20B via Groq, and AraBERT-ArSAS. The results show that sentiment predictions are not fully stable under paraphrasing. Gemini achieved the highest consistency rate at 90.67%, GPT-OSS-20B achieved the strongest overall accuracy and macro-F1, and AraBERT-ArSAS showed the highest flip rate. Error analysis showed that neutral-news confusion, dialect ambiguity, and sarcasm were the most common causes of sentiment flips. An ablation experiment with few-shot prompting improved GPT-OSS consistency from 85.33% to 90.00%, although the improvement was not statistically significant. These findings suggest that Arabic sentiment robustness should be evaluated not only by classification accuracy, but also by prediction stability under meaning-preserving text variations.**

***Index Terms*—Arabic NLP, sentiment analysis, paraphrasing, robustness, Arabic dialects, sarcasm detection, AraBERT, large language models**

I. Introduction

Arabic sentiment analysis is challenging because social media text often includes dialects, informal spelling, sarcasm, emojis, and short context. Standard metrics such as accuracy and F1-score show how well a model predicts labels, but they do not show whether the model is stable when the same meaning is expressed differently.

This project studies Arabic sentiment stability under paraphrasing. Using a combined dataset from ASTD and ArSarcasm, we test whether sentiment predictions remain consistent when Arabic tweets are paraphrased. A balanced sample of 150 tweets was paraphrased using Gemini 2.5 Flash, then classified by Gemini 2.5 Flash, GPT-OSS-20B via Groq, and AraBERT-ArSAS.

II. Dataset

This project uses two publicly available Arabic Twitter datasets: the Arabic Sentiment Tweets Dataset (ASTD) and ArSarcasm. These datasets were selected because the project studies sentiment stability under paraphrasing, which requires Arabic text with existing sentiment labels. ASTD provides general Arabic sentiment annotations, while ArSarcasm adds useful metadata such as sarcasm and dialect labels. This makes the combined dataset suitable for analyzing.

A. Data Sources

The first dataset is ASTD, introduced by Nabil, Aly, and Atiya as an Arabic social sentiment dataset collected from Twitter and annotated for sentiment analysis [1].

The second dataset is ArSarcasm, introduced by Abu Farha and Magdy for Arabic sarcasm and sentiment analysis [2]. ArSarcasm contains Arabic tweets with sentiment, sarcasm, dialect, source, and split information.

Together, the combined processed dataset contains 20,241 Arabic tweets.

B. Annotation Guidelines and Label Definitions

Both selected datasets already include sentiment annotations. Clear label definitions were documented to ensure consistent interpretation of the sentiment classes used in the project.

The original labels were mapped into three unified classes: `positive`, `negative`, and `neutral`. A tweet is labeled `positive` if it expresses praise, satisfaction, happiness, support, approval, or a favorable opinion. A tweet is labeled `negative` if it expresses criticism, anger, sadness, complaint, rejection, dissatisfaction, or an unfavorable opinion. A tweet is labeled `neutral` if it reports information, asks a question, describes an event, or does not clearly express positive or negative sentiment.

Table I shows examples of the annotation guidelines used to interpret the unified labels.

C. Preprocessing and Label Mapping

A preprocessing script was created to merge ASTD and ArSarcasm into a single dataset. It then standardizes column names, removes unnecessary surrounding quotation marks,

TABLE I
EXAMPLE SENTIMENT ANNOTATIONS

Example Text	Label	Reason
أحب هذا المنتج جدا	Positive	Expresses approval and satisfaction.
الخدمة سيئة جدا	Negative	Expresses dissatisfaction and criticism.
تم افتتاح الفرع الجديد اليوم	Neutral	Reports factual information.
ما هي مميزات درجة رجال الأعمال؟	Neutral	Asks a question without clear sentiment.

normalizes repeated whitespace, maps original labels to unified labels, adds metadata fields, removes missing rows, and saves the final dataset.

The final dataset schema includes: `id`, `text`, `sentiment_original`, `sentiment_unified`, `dataset`, `dialect`, `sarcasm`, `source`, and `split`.

The label mapping is shown in Table II. The ASTD label `OBJ` was mapped to `neutral` because objective tweets usually do not express clear positive or negative sentiment.

TABLE II
SENTIMENT LABEL MAPPING

Original Label	Dataset	Unified Label
POS	ASTD	positive
NEG	ASTD	negative
NEUTRAL	ASTD	neutral
OBJ	ASTD	neutral
positive	ArSarcasm	positive
negative	ArSarcasm	negative
neutral	ArSarcasm	neutral

D. Descriptive Statistics

Table III summarizes the two datasets used in this project. The combined dataset contains 20,241 tweets, with 10,547 tweets from ArSarcasm and 9,694 tweets from ASTD.

TABLE III
DATASET SOURCE SUMMARY

Dataset	Rows	Available Metadata
ASTD	9,694	Sentiment
ArSarcasm	10,547	Sentiment, sarcasm, dialect, source
Total	20,241	Combined metadata

The unified sentiment distribution is shown in Table IV. The dataset is imbalanced, with the neutral class forming the largest portion of the data.

Dialect information is available mainly from ArSarcasm. Since ASTD does not provide dialect labels, its rows were assigned the value `unknown`. Table V shows the dialect distribution in the combined dataset.

TABLE IV
UNIFIED SENTIMENT DISTRIBUTION

Sentiment	Count	Percentage
Neutral	12,615	62.33%
Negative	5,171	25.55%
Positive	2,455	12.13%

TABLE V
DIALECT DISTRIBUTION IN THE COMBINED DATASET

Dialect	Count	Percentage
Unknown	9,694	47.89%
MSA	7,062	34.89%
Egyptian	2,383	11.77%
Levantine	551	2.72%
Gulf	519	2.56%
Maghrebi	32	0.16%

E. Inter-Annotator Agreement

Since both datasets already provide sentiment labels, the original labels were used as reference annotations.

F. Biases and Limitations

There are several limitations in the prepared dataset. First, both datasets are based on Twitter data, so the language is short, informal, and affected by social media writing conventions such as hashtags, mentions, spelling variation, and slang. Second, the dataset is class-imbalanced, with neutral tweets representing 62.33% of the combined data. Third, dialect metadata is unavailable for ASTD, so dialect-based analysis depends mostly on ArSarcasm. Fourth, the dialect distribution is also imbalanced: MSA and Egyptian Arabic are much more frequent than Gulf, Levantine, and Maghrebi Arabic. Finally, mapping ASTD's `OBJ` label to `neutral` simplifies the original label scheme and may hide differences between objective factual tweets and neutral subjective tweets.

III. Experimental Setup and Results

This section reports the baseline experiments for testing Arabic sentiment stability under paraphrasing. The goal is to evaluate whether different sentiment models produce stable predictions when the input tweet is paraphrased while preserving its meaning.

A. Research Questions and Hypotheses

The main research question is: do Arabic sentiment models maintain the same sentiment label after paraphrasing? This was formalized into the following testable hypotheses:

- **H1:** Arabic sentiment predictions will change for at least 10% of paraphrased tweets.
- **H2:** The three tested models will differ in their sentiment consistency rates.
- **H3:** Sarcastic tweets will show lower consistency than non-sarcastic tweets.
- **H4:** Dialectal Arabic tweets will show more sentiment drift than Modern Standard Arabic (MSA) tweets.

B. Experimental Data

The experiment used a balanced subset of 150 tweets sampled from the processed dataset described in the previous section. The sample contained 50 positive, 50 negative, and 50 neutral tweets.

C. Paraphrase Generation

Each original tweet was paraphrased once using Gemini 2.5 Flash. The paraphrasing prompt instructed the model to preserve the original meaning, sentiment, and dialect as much as possible, and to return only the paraphrased Arabic text. The paraphrasing temperature was set to 0.7 to allow natural variation.

The paraphrase prompt template was:

> Paraphrase the following Arabic tweet while preserving its original meaning and sentiment. Keep the same sentiment, keep the same dialect as much as possible, do not add new information, and return only the paraphrased Arabic text.

Using one shared paraphrase source made the comparison between sentiment models more controlled. All three models were evaluated on the same original–paraphrase pairs.

D. Models

Three models were compared. Gemini 2.5 Flash was used as a general instruction-following language model. GPT-OSS-20B hosted through Groq was used as an open GPT-style model [3]. AraBERT-ArSAS was used as the Arabic-specific transformer baseline. Its Hugging Face model card describes it as an AraBERT-v2 model fine-tuned on the ArSAS dataset for sentiment classification with the labels negative, neutral, positive, and mixed [4].

For Gemini and GPT-OSS, the classification task was performed using a zero-shot prompt. The prompt asked the model to classify each Arabic text into exactly one label: `positive`, `negative`, or `neutral`. The classification temperature was set to 0.0 to reduce output variation. AraBERT-ArSAS was run locally using the Hugging Face Transformers pipeline.

E. Evaluation Metrics

The experiment used both correctness-based and stability-based metrics. Accuracy and macro-F1 were computed by comparing model predictions with the gold sentiment labels from the dataset. These metrics were computed separately for original tweets and paraphrased tweets.

The main robustness metric was consistency rate:

$$Consistency = \frac{\#\text{ unchanged predictions}}{\#\text{ original-paraphrase pairs}} \tag{1}$$

A prediction was considered consistent when the model assigned the same label to the original tweet and its paraphrase. The flip rate was calculated as:

$$FlipRate = 1 - Consistency \tag{2}$$

The analysis also included per-class consistency, per-dialect consistency, and sarcasm-based consistency.

F. Aggregate Results

Table VI shows the aggregate results for the three models. Gemini achieved the highest consistency rate at 90.67%, followed by GPT-OSS-20B via Groq at 85.33%, and AraBERT-ArSAS at 78.52%. This supports H1 and H2: all models showed some sentiment drift after paraphrasing, and the models differed in their stability.

The results show that accuracy and stability are not identical. GPT-OSS-20B achieved the best overall accuracy and macro-F1, while Gemini produced the most stable predictions after paraphrasing. AraBERT had the lowest consistency, but its accuracy improved after paraphrasing, suggesting that some paraphrases may have made informal or noisy tweets easier for the Arabic transformer model to classify.

G. Per-Class Results

Table VII shows consistency by gold sentiment class. Gemini was the most stable across all three classes, with consistency above 90% for positive tweets and 90% for both negative and neutral tweets. GPT-OSS-20B showed moderate stability, while AraBERT was less stable, especially for positive tweets.

The per-class F1 scores showed that GPT-OSS-20B had the strongest overall classification performance. Gemini had strong performance on positive and negative examples, but its neutral recall was low, indicating that it often interpreted neutral tweets as emotional. AraBERT improved after paraphrasing across all classes, which may indicate that the paraphrases reduced spelling noise, dialectal variation, or informal phrasing.

H. Dialect and Sarcasm Analysis

Dialect-level analysis showed mixed patterns. For Gemini, MSA tweets had a consistency rate of 91.94%, Egyptian tweets had 85.71%, and unknown-dialect tweets had 89.47%. GPT-OSS-20B showed 87.10% consistency on MSA and 80.95% on Egyptian tweets, but only 57.14% on Gulf tweets. AraBERT showed lower consistency on Egyptian and Levantine tweets, both at 66.67%, compared with 85.96% for unknown-dialect tweets.

These results provide partial support for H4, but the dialect sample sizes were uneven.

For sarcasm, the results did not consistently support H3. GPT-OSS-20B showed lower consistency on sarcastic tweets than non-sarcastic tweets, with 75.00% consistency for sarcastic tweets compared with 84.42% for non-sarcastic tweets. However, Gemini showed high consistency for sarcastic tweets at 93.75%, and AraBERT showed 80.00% consistency for sarcastic tweets compared with 72.73% for non-sarcastic tweets. This suggests that sarcasm effects require a larger sample before stronger conclusions can be made.

I. Error Analysis

A total of 68 sentiment flips were found across the three models. Table **??** summarizes representative failure cases and their likely causes.

TABLE VI
AGGREGATE EXPERIMENTAL RESULTS

Model	Orig. Acc.	Para. Acc.	Orig. Macro-F1	Para. Macro-F1	Consistency	Flip Rate
AraBERT-ArSAS	61.74%	67.33%	61.68%	66.72%	78.52%	21.48%
GPT-OSS-20B via Groq	73.33%	73.33%	72.96%	72.76%	85.33%	14.67%
Gemini 2.5 Flash	69.33%	67.33%	66.63%	65.26%	90.67%	9.33%

TABLE VII
PER-CLASS CONSISTENCY RATES

Model	Negative	Neutral	Positive
AraBERT-ArSAS	85.71%	78.00%	72.00%
GPT-OSS-20B via Groq	84.00%	84.00%	88.00%
Gemini 2.5 Flash	90.00%	90.00%	92.00%

J. Summary of Findings

Overall, the baseline experiment shows that Arabic sentiment prediction is not fully stable under paraphrasing. Gemini was the most consistent model, GPT-OSS-20B via Groq achieved the strongest accuracy and macro-F1, and AraBERT-ArSAS was the least stable but improved after paraphrasing. The findings suggest that paraphrasing can change model behavior even when the intended meaning is preserved.

IV. Analysis and Discussion

This section extends the baseline results by analyzing the failure cases in more detail.

A. Expanded Error Analysis

The baseline experiment produced 68 sentiment flip cases, where a model assigned different labels to the original tweet and its paraphrase. These cases were grouped into systematic error categories.

Table IX summarizes the main error categories. The most frequent category was neutral-news confusion.

Neutral-news confusion accounted for 27 out of 68 flips. This suggests that models sometimes rely on topic-level emotional association rather than the actual sentiment expressed by the author.

Dialect and slang ambiguity accounted for 17 flips. This supports the observation that Arabic sentiment robustness is affected also by dialectal wording.

B. Ablation Experiment: Few-Shot Prompting

The expanded error analysis showed that many flips came from neutral-news confusion and ambiguous social media expressions. To test whether prompt design could reduce instability, an ablation experiment was conducted on GPT-OSS-20B via Groq.

The baseline used a zero-shot sentiment classification prompt. The ablation used a few-shot prompt with label definitions and examples for positive, negative, and neutral classes. The examples were selected to address common baseline errors, including neutral factual headlines, sports-related sentiment, and sarcastic negative wording.

Table X compares GPT-OSS zero-shot and few-shot prompting.

The few-shot prompt did not change accuracy, but it improved consistency from 85.33% to 90.00%. It also reduced the flip rate from 14.67% to 10.00%. This suggests that examples can make the model's decision boundary more stable even when they do not improve correctness against the gold label.

Table XI shows that the largest improvement occurred for neutral tweets. Neutral consistency increased from 84.00% to 92.00%. This directly connects to the error analysis, where neutral-news confusion was the most frequent error category.

C. Statistical Significance Testing

McNemar's test was used to test whether differences in consistency were statistically significant. The test was applied to paired consistency outcomes, where each tweet pair was marked as either consistent or flipped for each model or configuration.

Table XII reports the p-values.

The difference between Gemini and AraBERT was statistically significant at the 0.05 level, indicating that Gemini was significantly more stable than AraBERT on this sample. However, the difference between Gemini and GPT-OSS zero-shot was not statistically significant. Similarly, the few-shot ablation improved GPT-OSS consistency numerically, but the difference was not statistically significant at the 0.05 level.

D. Connection to Related Work

Our findings agree with that Arabic models remain sensitive to small input changes. Radman and Duwairi studied robustness in Arabic sentiment analysis using adversarial synonym replacement and adversarial training [6]. Our work similarly finds that meaning-preserving changes can still cause sentiment flips.

The error analysis also supports prior findings that dialectal Arabic is difficult for sentiment models. Alshahrani evaluated Arabic adversarial examples using both MSA and dialectal Arabic datasets [7]. In our results, dialect and slang ambiguity formed 25.00% of flip cases.

Finally, Li et al. showed that paraphrases can help connect implicit and explicit sentiment expressions [8]. Our findings support this idea, but also show that paraphrasing can clarify sentiment in some cases.

E. Limitations

Paraphrases were generated by one model, Gemini 2.5 Flash. This made the model comparison more controlled, but it also means that the results depend on one paraphrase style.

TABLE VIII
REPRESENTATIVE SENTIMENT FLIP CASES

Model	Original Text	Paraphrase	Orig. Pred.	Para. Pred.	Possible Explanation
Gemini	مصر دولة بس مش أوي	مصر دولة على الورق، لكن في الحقيقة هي مش كده.	negative	positive	Sarcasm ambiguity.
Gemini	صورة: كاسياس يحتفل بالانتصار في ديربي مدريد	تُظهر هذه الصورة كاسياس وهو يحتفل بالانتصار في ديربي مدريد.	positive	neutral	The paraphrase became more descriptive and reduced the emotional signal.
Gemini	مقتل سعيد فضة أكبر مورد بلطجية فى شبرا الخيمة وعميل أمن الدولة	تأكيد مقتل سعيد فضة، المعروف بكونه أكبر مورد للبلطجية في شبرا الخيمة، وعميل جهاز أمن الدولة.	negative	positive	News and entity wording confused the model.
GPT-OSS	استشهاد جنود مصريين على الحدود بنيران صهيونية جريمة غير مقبولة	استشهاد جنودنا المصريين على الحدود بنيران صهيونية عمل إجرامي مرفوض تماماً.	negative	positive	Patriotic wording may have introduced positive cues
GPT-OSS	الارجنتين بدون ميسي مضحكه	الأرجنتين بلا ميسي مسخرة.	positive	negative	Sarcasm and ambiguity.
GPT-OSS	تكريم رامي عياش بجامعة ناو في لبنان	رامي عياش يُكرّم في جامعة ناو اللبنانية.	positive	neutral	We are not sure why the flip happened.
AraBERT	عجيييييب	خرافي!	positive	negative	AraBERT misclassified the short informal paraphrase.
AraBERT	افتكرت تعليق علاء الحامولي ع اللي حصل لعادل الأمور في ماتش المنيا زمان	افتكرت تعليق علاء الحامولي على اللي حصل لعادل الأمور في ماتش المنيا من زمن بعيد أوي.	positive	negative	Dialect and informal context ambiguity.

TABLE IX
EXPANDED ERROR CATEGORY SUMMARY

Error Category	Count	Percentage
Neutral-news confusion	27	39.71%
Dialect/slang ambiguity	17	25.00%
Sarcasm/irony ambiguity	13	19.12%
Paraphrase clarified sentiment	6	8.82%
Short text/context loss	2	2.94%
Entity/topic bias	1	1.47%
Paraphrase sentiment shift	1	1.47%
Gold-label/context ambiguity	1	1.47%

Also, the error categories were assigned using a rule-based analysis informed by manual inspection.

F. Exploratory Observations and Future Work

One exploratory observation is that paraphrasing sometimes improved classification, especially for AraBERT.

Another observation is that neutral tweets were especially fragile. Neutral-news confusion was the most frequent error type, and the few-shot ablation improved neutral consistency the most. Future experiments could focus specifically on separating neutral factual reporting from emotionally loaded topics.

Future work should also test multiple paraphrases per tweet instead of one. This would allow measuring whether a model is stable across a range of semantically similar alternatives.

V. CONCLUSION

This project evaluated Arabic sentiment stability under paraphrasing using tweets from ASTD and ArSarcasm. Three models were compared: Gemini 2.5 Flash, GPT-OSS-20B via Groq, and AraBERT-ArSAS. The results showed that sentiment predictions are not fully stable after paraphrasing. Gemini had the highest consistency rate, GPT-OSS achieved the best accuracy and macro-F1, and AraBERT showed the highest flip rate.

Error analysis showed that most flips were caused by neutral-news confusion, dialect ambiguity, and sarcasm. A few-shot prompting ablation improved GPT-OSS consistency, especially for neutral tweets, but the improvement was not statistically significant. Overall, the findings show that Arabic sentiment models should be evaluated using robustness measures such as consistency rate in addition to traditional accuracy-based metrics.

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TABLE X
GPT-OSS PROMPTING ABLATION RESULTS

Configuration	Orig. Acc.	Para. Acc.	Orig. Macro-F1	Para. Macro-F1	Consistency	Flip Rate
Zero-shot	73.33%	73.33%	72.96%	72.76%	85.33%	14.67%
Few-shot	73.33%	73.33%	73.15%	73.21%	90.00%	10.00%

TABLE XI
GPT-OSS PER-CLASS CONSISTENCY IN THE ABLATION

Configuration	Negative	Neutral	Positive
Zero-shot	84.00%	84.00%	88.00%
Few-shot	88.00%	92.00%	90.00%

TABLE XII
MCNEMAR'S TEST ON CONSISTENCY DIFFERENCES

Comparison	p-value	Significant?
Gemini vs GPT-OSS zero-shot	0.2005	No
Gemini vs AraBERT	0.0066	Yes
GPT-OSS zero-shot vs few-shot	0.1185	No

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APPENDIX A
PROJECT REPOSITORY

The project repository contains the dataset preparation scripts, experiment scripts, prompt templates, ablation results, statistical analysis scripts, and result files used in this study. GitHub repository: `des3bd/arabic-sentiment-stability`